Long-range cutoff: 6 Å vs 12 Å distance scans

Two charge/spin-aware MACE models were trained from scratch on the same 20k OMol25 frames: SPICE-2 elements, $|q| \leq 2$, 15 epochs, energy:force 1:1, fixed ωB97M-V atom E0s. They are identical apart from the cutoff (r_max 6 or 12 Å). Their validation energy RMSEs are 97.5 meV/atom (6 Å) and 88.3 (12 Å).

Eleven fragment pairs were pulled apart rigidly from 2.5 to 20 Å minimum gap. ORCA ωB97M-V/def2-TZVPD (RIJCOSX, def2/J, TightSCF, DEFGRID3) was run at every point, 195 single points in all.

Method. $\Delta E(d) = E(\text{pair}) - E(A) - E(B)$, in kcal/mol. Tables give each curve relative to its own value at 20 Å, so only the shape is compared; this removes the models' constant charge-state offset. Raw ΔE at 20 Å is quoted in the notes. Gap is the minimum atom–atom distance between the fragments.

Result. The 12 Å model's curves change out to about 10 Å; the 6 Å model's go flat at 5–6 Å, for every pair. The extra range gets close to the DFT shape for cation–water and cation–π. It doesn't learn the 1/r tail of ion pairs, and gets the sign wrong for the cation–cation pair. Between neutral molecules it adds spurious wells at 5–10 Å. Ion–ion behaviour likely needs explicit long-range electrostatics (e.g. MACE-POLAR), possibly with a 12 Å or longer cutoff for the cation–ligand gains.

Summary

Table 1

Pair	Interaction	DFT 3 / 8 Å	6 Å 3 / 8 Å	12 Å 3 / 8 Å	12 Å vs 6 Å
NH_4^+ · acetate	attractive ion pair, r^{-1}	−76.51 / −21.51	−7.34 / +0.00	−8.81 / −1.62	Slight gain
NH_4^+ · Cl^-	attractive ion pair, r^{-1}	−74.75 / −21.22	−2.32 / +0.00	−1.47 / −0.19	No gain
NH_4^+ · NH_4^+	repulsive ion pair, r^{-1}	+51.05 / +18.00	−5.02 / +0.00	−30.06 / −1.54	12 Å worse
Na^+ · water	ion–dipole, r^{-2}	−15.90 / −1.72	−7.54 / +0.00	−11.53 / −1.13	12 Å better
Cl^- · water	ion–dipole, r^{-2}	−11.46 / −1.51	−4.53 / +0.00	−2.88 / −0.29	No gain
Na^+ · methane	ion–induced dipole, r^{-4}	−3.31 / −0.08	+1.36 / +0.00	−1.99 / −0.97	Mixed
Na^+ · benzene	cation–π and induction	−24.17 / −1.20	−9.14 / +0.00	−9.06 / −3.90	12 Å better
Water dimer, aligned dipoles	dipole–dipole, r^{-3}	−2.02 / −0.14	−2.03 / +0.00	−2.50 / −1.12	12 Å worse
Acetonitrile dimer, head to tail	dipole–dipole, r^{-3}	−3.04 / −0.34	−3.61 / +0.00	−10.47 / −1.60	12 Å worse
Methane dimer	dispersion, r^{-6}	−0.23 / −0.00	−1.16 / +0.00	−0.64 / −0.83	12 Å worse
Benzene dimer, stacked	dispersion and quadrupole	+11.68 / +0.01	+10.96 / +0.00	+1.75 / −7.29	12 Å worse

Ion–ion

NH_4^+ · acetate (attractive ion pair, r^{-1}): Slight gain

DFT has the long Coulomb tail: −76.5 kcal/mol at 3 Å and still −6.4 at 14 Å. Neither model follows it. The 6 Å curve is flat past 6 Å. The 12 Å curve keeps a small slope out to 10 Å and recovers about 10% of the DFT change. In raw ΔE both models put most of the attraction into a constant: −17.2 (6 Å) and −25.0 (12 Å) at 20 Å, against DFT −15.9.

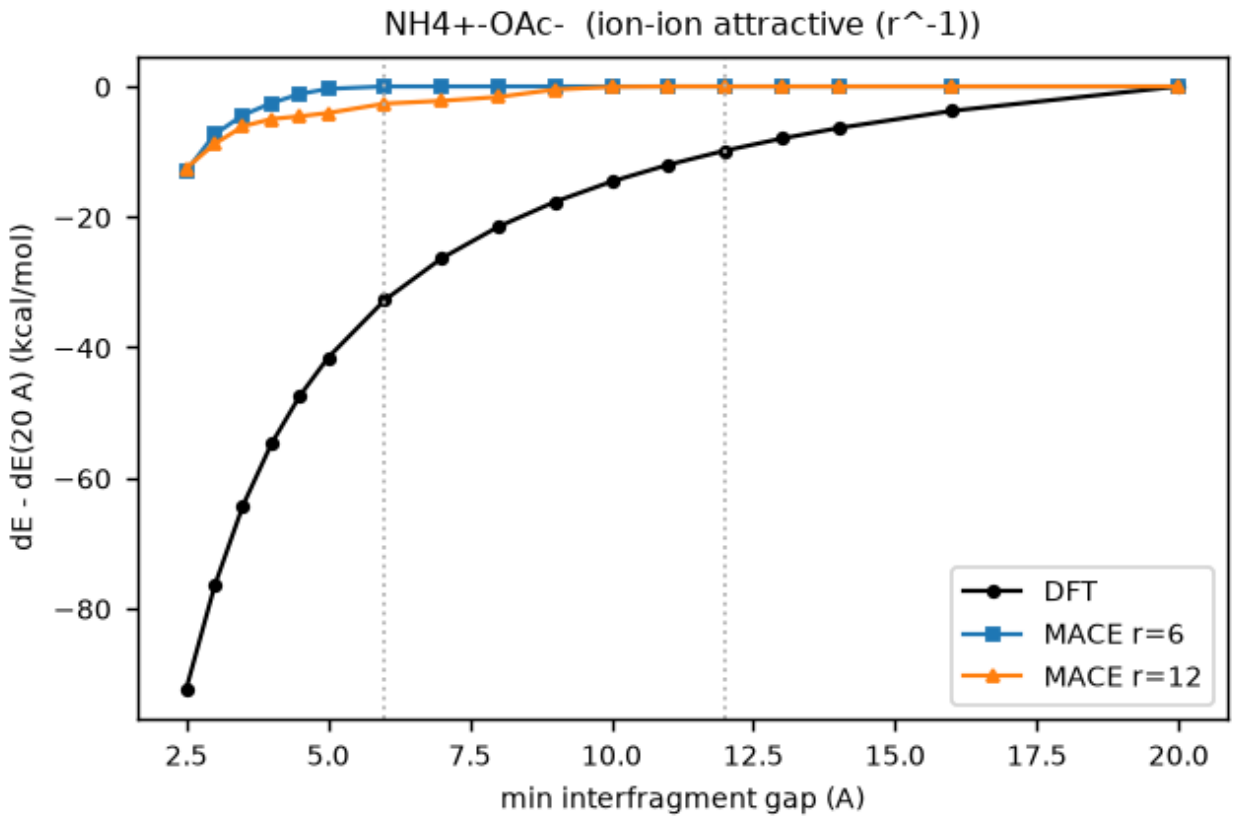


Table 2

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−92.29	−12.98	−12.53	+79.76
3.00	−76.51	−7.34	−8.81	+67.70
3.50	−64.20	−4.39	−6.11	+58.09
4.00	−54.78	−2.56	−5.00	+49.79
4.50	−47.44	−1.16	−4.58	+42.86
5.00	−41.56	−0.37	−4.07	+37.49
6.00	−32.73	+0.00	−2.63	+30.10
7.00	−26.36	+0.00	−2.21	+24.15
8.00	−21.51	+0.00	−1.62	+19.89
9.00	−17.68	+0.00	−0.48	+17.20
10.00	−14.57	+0.00	−0.06	+14.51

11.00	−12.00	+0.00	+0.01	+12.01
12.00	−9.84	+0.00	+0.00	+9.84
13.00	−7.98	+0.00	+0.00	+7.98
14.00	−6.38	+0.00	+0.00	+6.38
16.00	−3.75	+0.00	+0.00	+3.75
20.00	+0.00	+0.00	+0.00	+0.00

$NH_4^+ \cdot Cl^-$ (attractive ion pair, r^{-1}): No gain

The DFT tail is the same as for the acetate pair. Both models are almost flat, with under 2% of the DFT change at any gap. Raw ΔE at 20 Å is −34.3 (6 Å) and −54.7 (12 Å), against DFT −15.8.

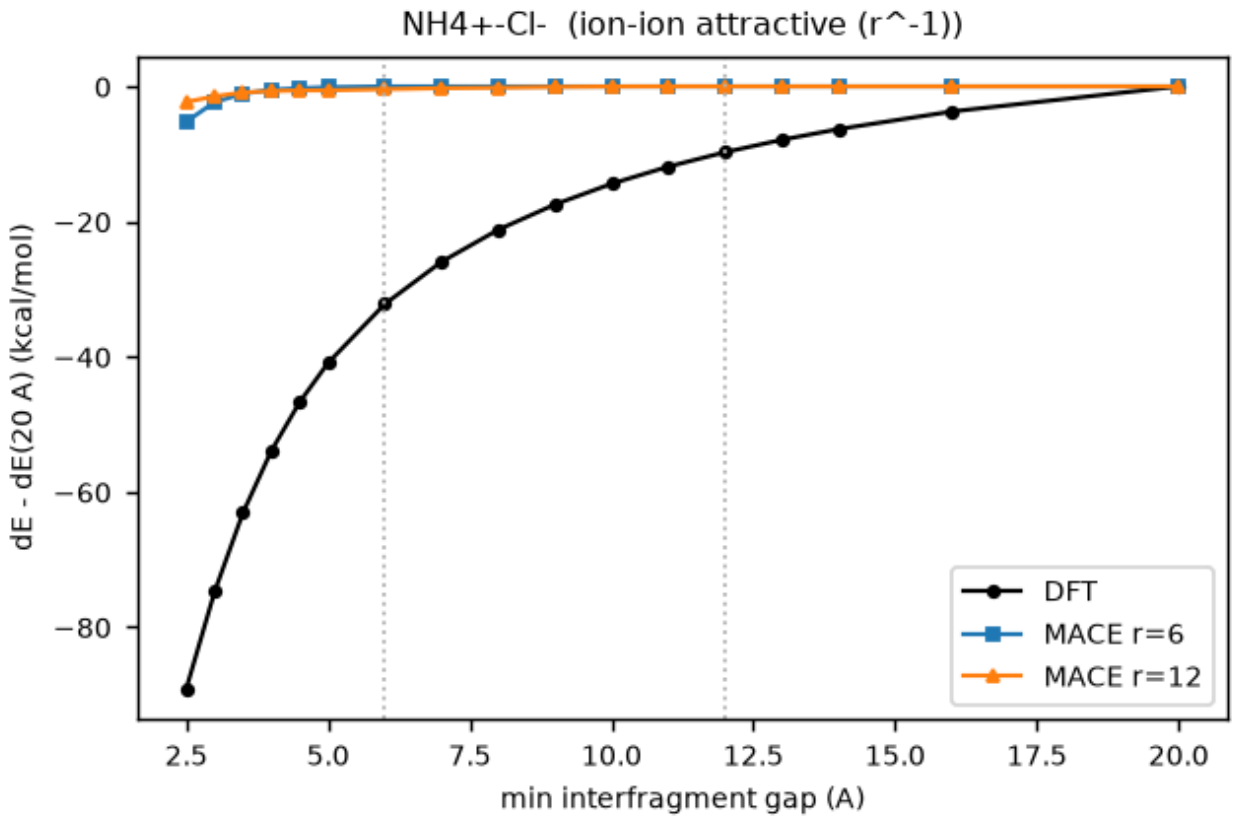


Table 3

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−89.12	−5.22	−2.31	+86.81
3.00	−74.75	−2.32	−1.47	+73.27
3.50	−63.08	−1.08	−0.91	+62.17
4.00	−53.92	−0.49	−0.68	+53.24
4.50	−46.70	−0.27	−0.59	+46.12
5.00	−40.91	−0.12	−0.57	+40.34
6.00	−32.21	+0.00	−0.42	+31.79

7.00	−25.96	+0.00	−0.26	+25.70
8.00	−21.22	+0.00	−0.19	+21.03
9.00	−17.47	+0.00	−0.08	+17.40
10.00	−14.41	+0.00	−0.01	+14.40
11.00	−11.87	+0.00	+0.00	+11.88
12.00	−9.74	+0.00	+0.00	+9.74
13.00	−7.91	+0.00	+0.00	+7.91
14.00	−6.32	+0.00	+0.00	+6.32
16.00	−3.72	+0.00	+0.00	+3.72
20.00	+0.00	+0.00	+0.00	+0.00

$NH_4^+ \cdot NH_4^+$ (repulsive ion pair, r^{-1}): 12 Å worse

DFT is repulsive: +51 kcal/mol at 3 Å, decaying slowly. The 6 Å model is nearly flat. The 12 Å model has the wrong sign: about −30 at 3 Å and −12 at 6 Å, so it pulls the two cations together. Raw ΔE at 20 Å is +131 for 12 Å, against +44 for 6 Å and +15 for DFT.

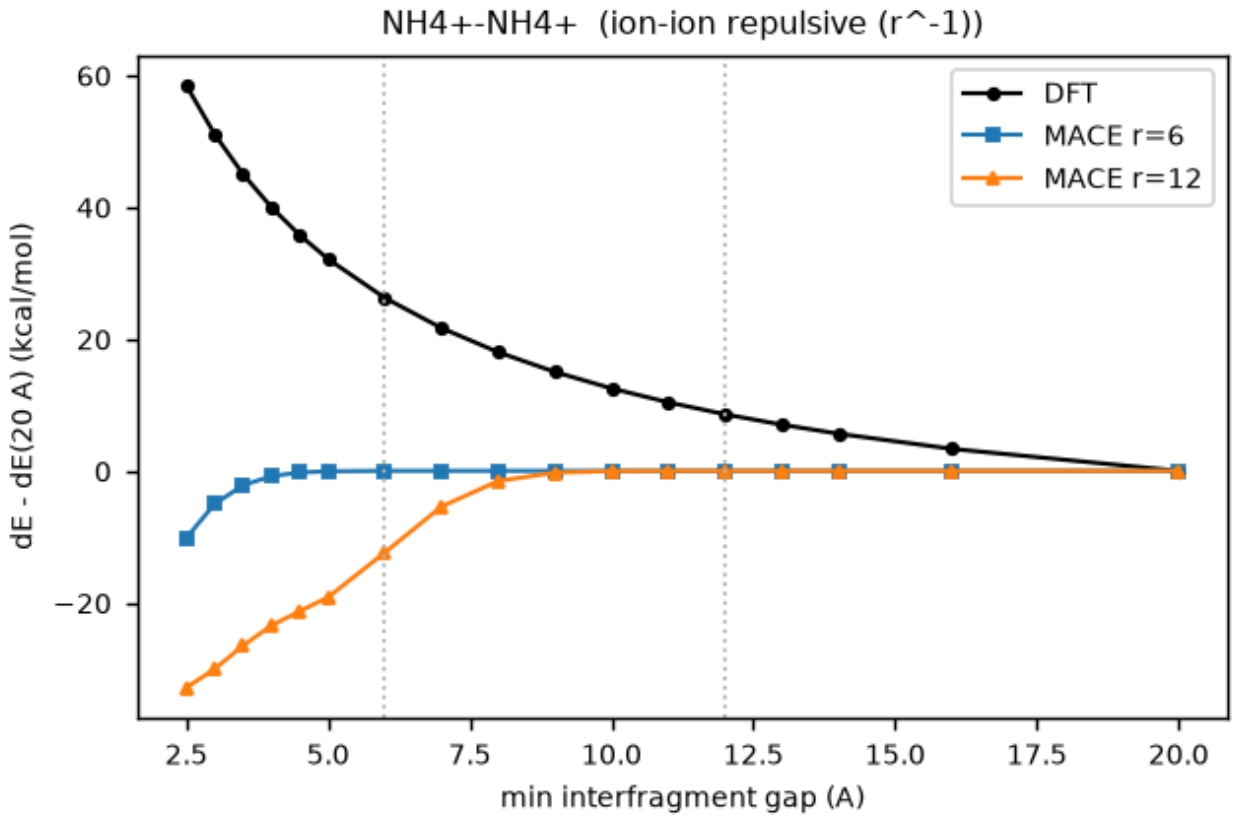


Table 4

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	+58.47	−10.17	−32.88	−91.35
3.00	+51.05	−5.02	−30.06	−81.11

3.50	+45.01	−2.19	−26.52	−71.53
4.00	+39.99	−0.81	−23.48	−63.47
4.50	+35.76	−0.26	−21.29	−57.05
5.00	+32.13	−0.07	−19.17	−51.30
6.00	+26.24	+0.00	−12.40	−38.64
7.00	+21.66	+0.00	−5.42	−27.08
8.00	+18.00	+0.00	−1.54	−19.54
9.00	+15.00	+0.00	−0.27	−15.27
10.00	+12.50	+0.00	−0.04	−12.54
11.00	+10.39	+0.00	−0.00	−10.39
12.00	+8.58	+0.00	+0.00	−8.58
13.00	+7.01	+0.00	+0.00	−7.01
14.00	+5.63	+0.00	+0.00	−5.63
16.00	+3.34	+0.00	+0.00	−3.34
20.00	+0.00	+0.00	+0.00	+0.00

Ion–neutral

Na^+ · water (ion–dipole, r^{-2}): 12 Å better

The 12 Å model is about twice as close to DFT as the 6 Å model at every gap out to 8 Å. At 4 Å it gives −4.8 against DFT −8.4 (6 Å gives −2.3). At 8 Å it gives −1.1 against −1.7, where the 6 Å model is already flat. Both still undershoot the well at 3 Å.

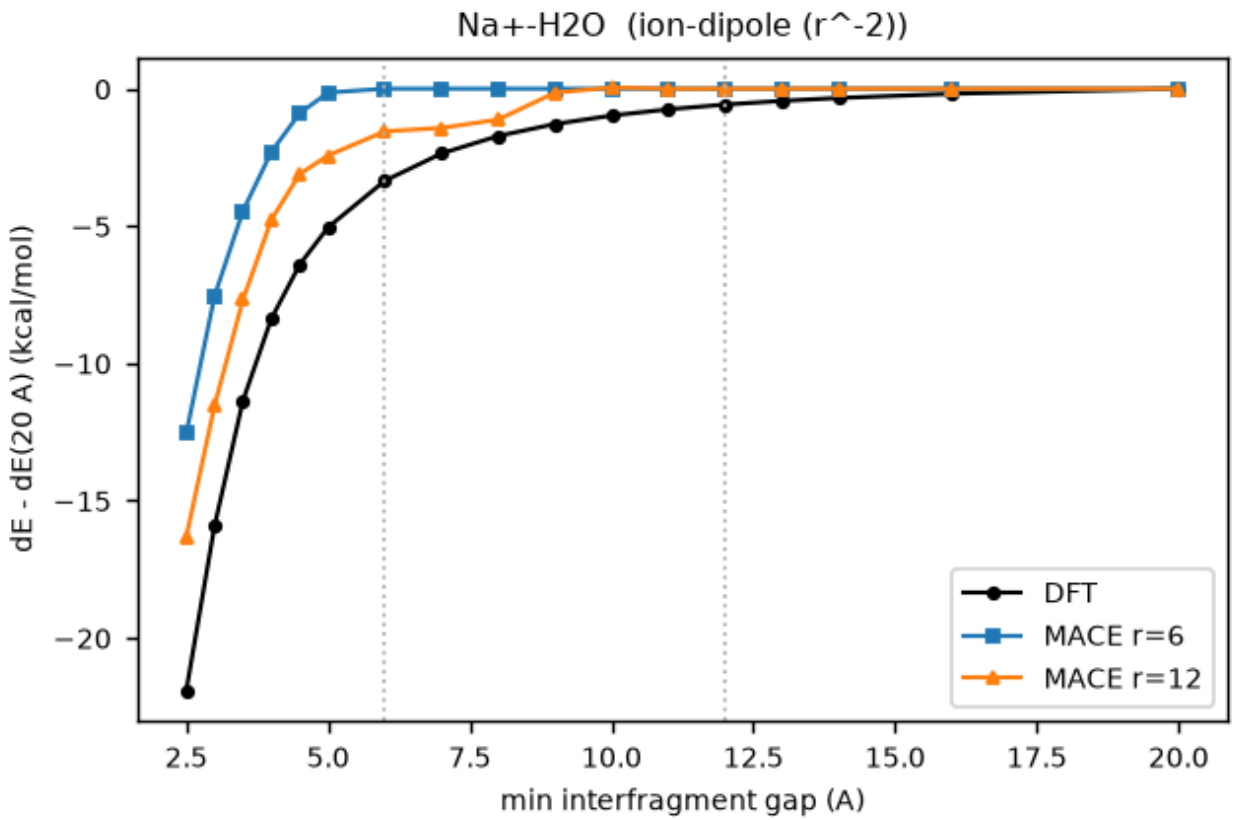


Table 5

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−21.92	−12.55	−16.32	+5.60
3.00	−15.90	−7.54	−11.53	+4.38
3.50	−11.35	−4.45	−7.68	+3.68
4.00	−8.38	−2.30	−4.78	+3.60
4.50	−6.40	−0.89	−3.10	+3.30
5.00	−5.05	−0.14	−2.44	+2.61
6.00	−3.35	+0.00	−1.55	+1.80
7.00	−2.36	+0.00	−1.43	+0.93
8.00	−1.72	+0.00	−1.13	+0.60
9.00	−1.29	+0.00	−0.14	+1.14
10.00	−0.98	+0.00	+0.04	+1.02
11.00	−0.75	+0.00	−0.00	+0.75
12.00	−0.58	+0.00	+0.00	+0.58
13.00	−0.45	+0.00	+0.00	+0.45
14.00	−0.34	+0.00	+0.00	+0.34
16.00	−0.18	+0.00	+0.00	+0.18

20.00	+0.00	+0.00	+0.00	+0.00

Cl^- · water (ion–dipole, r^{-2}): No gain

Neither model follows the anion–water attraction beyond contact. The 6 Å model is slightly better at 3 Å (−4.5 against DFT −11.5). The 12 Å model is flat from 4 Å.

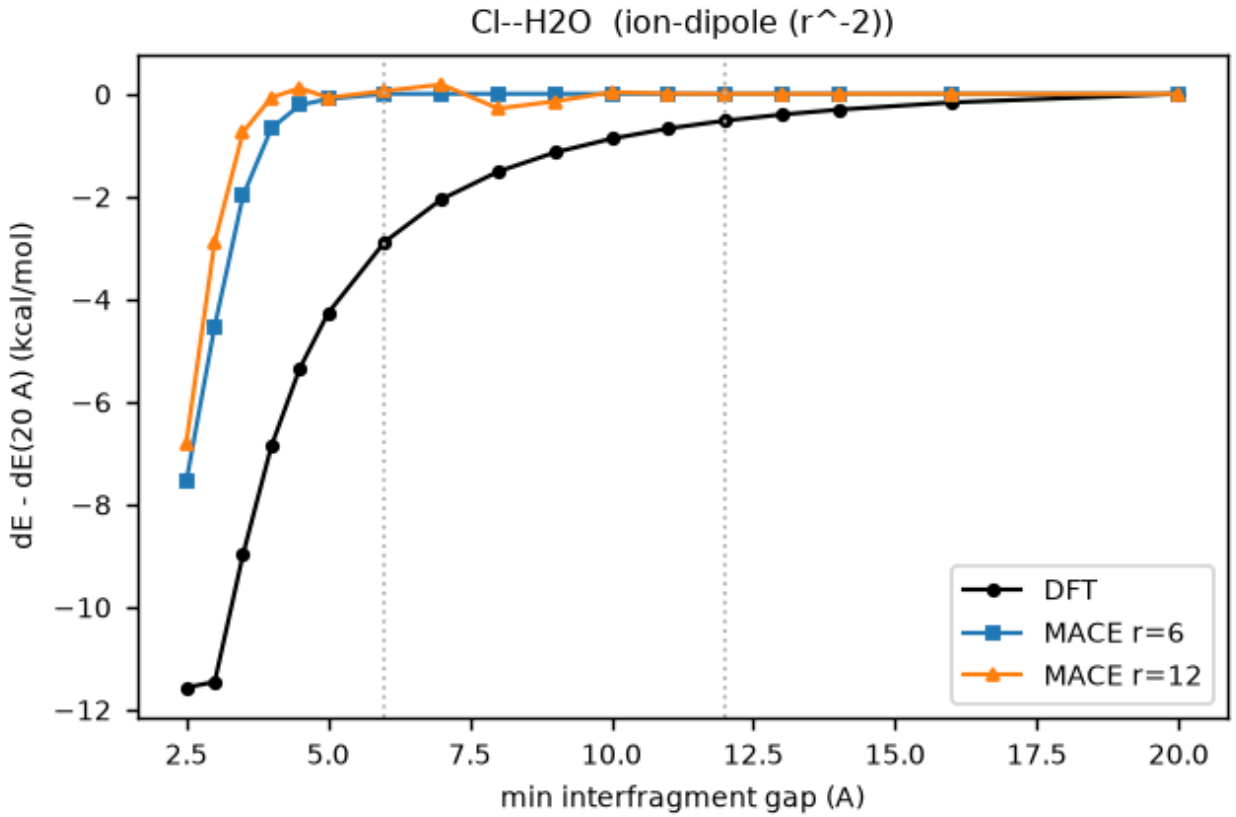


Table 6

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−11.57	−7.56	−6.80	+4.77
3.00	−11.46	−4.53	−2.88	+8.58
3.50	−8.98	−1.99	−0.75	+8.23
4.00	−6.87	−0.67	−0.07	+6.80
4.50	−5.36	−0.22	+0.11	+5.47
5.00	−4.28	−0.10	−0.07	+4.20
6.00	−2.89	+0.00	+0.06	+2.94
7.00	−2.06	+0.00	+0.18	+2.24
8.00	−1.51	+0.00	−0.29	+1.23
9.00	−1.14	+0.00	−0.15	+1.00
10.00	−0.88	+0.00	+0.03	+0.91
11.00	−0.68	+0.00	+0.01	+0.68

12.00	−0.52	+0.00	+0.00	+0.52
13.00	−0.41	+0.00	+0.00	+0.41
14.00	−0.31	+0.00	+0.00	+0.31
16.00	−0.17	+0.00	+0.00	+0.17
20.00	+0.00	+0.00	+0.00	+0.00

Na^+ · methane (ion–induced dipole, r^{-4}): Mixed

At 3 Å the 6 Å model has the wrong sign (+1.4 against DFT −3.3), and the 12 Å model is close (−2.0). From 4 to 8 Å the 12 Å model overshoots, holding about −2 kcal/mol where DFT has decayed to under −0.5.

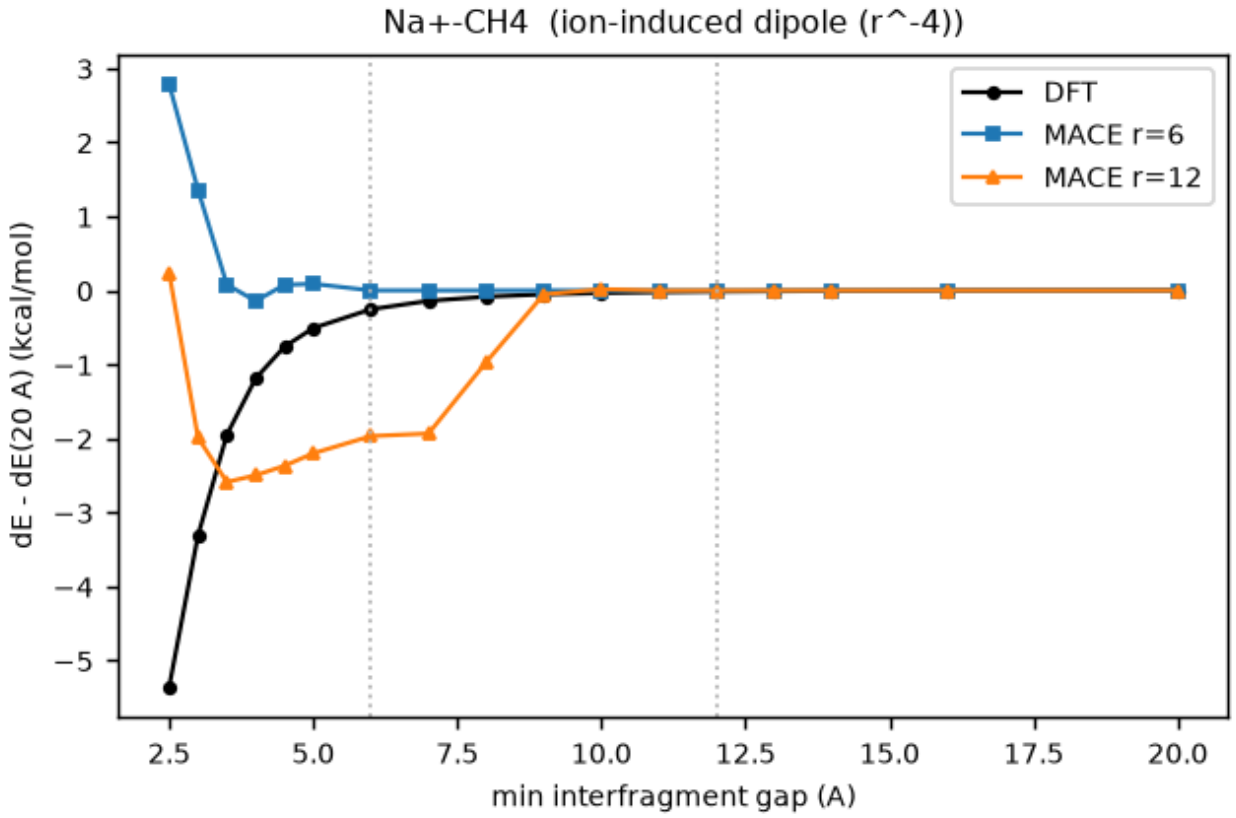


Table 7

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−5.35	+2.78	+0.23	+5.58
3.00	−3.31	+1.36	−1.99	+1.32
3.50	−1.95	+0.08	−2.59	−0.64
4.00	−1.19	−0.14	−2.50	−1.31
4.50	−0.76	+0.07	−2.37	−1.61
5.00	−0.51	+0.09	−2.20	−1.69
6.00	−0.25	+0.00	−1.97	−1.71
7.00	−0.14	+0.00	−1.93	−1.79

8.00	−0.08	+0.00	−0.97	−0.88
9.00	−0.05	+0.00	−0.05	+0.00
10.00	−0.04	+0.00	+0.02	+0.05
11.00	−0.02	+0.00	−0.00	+0.02
12.00	−0.02	+0.00	+0.00	+0.02
13.00	−0.01	+0.00	+0.00	+0.01
14.00	−0.01	+0.00	+0.00	+0.01
16.00	−0.00	+0.00	+0.00	+0.00
20.00	+0.00	+0.00	+0.00	+0.00

Na^+ · benzene (cation–π and induction): 12 Å better

The clearest gain. From 4 to 6 Å the 12 Å model tracks DFT (−9.5, −6.1, −4.1 against −11.5, −5.6, −3.1), while the 6 Å model is already near zero. It overshoots at 8 Å (−3.9 against −1.2) and drops to zero by 10 Å.

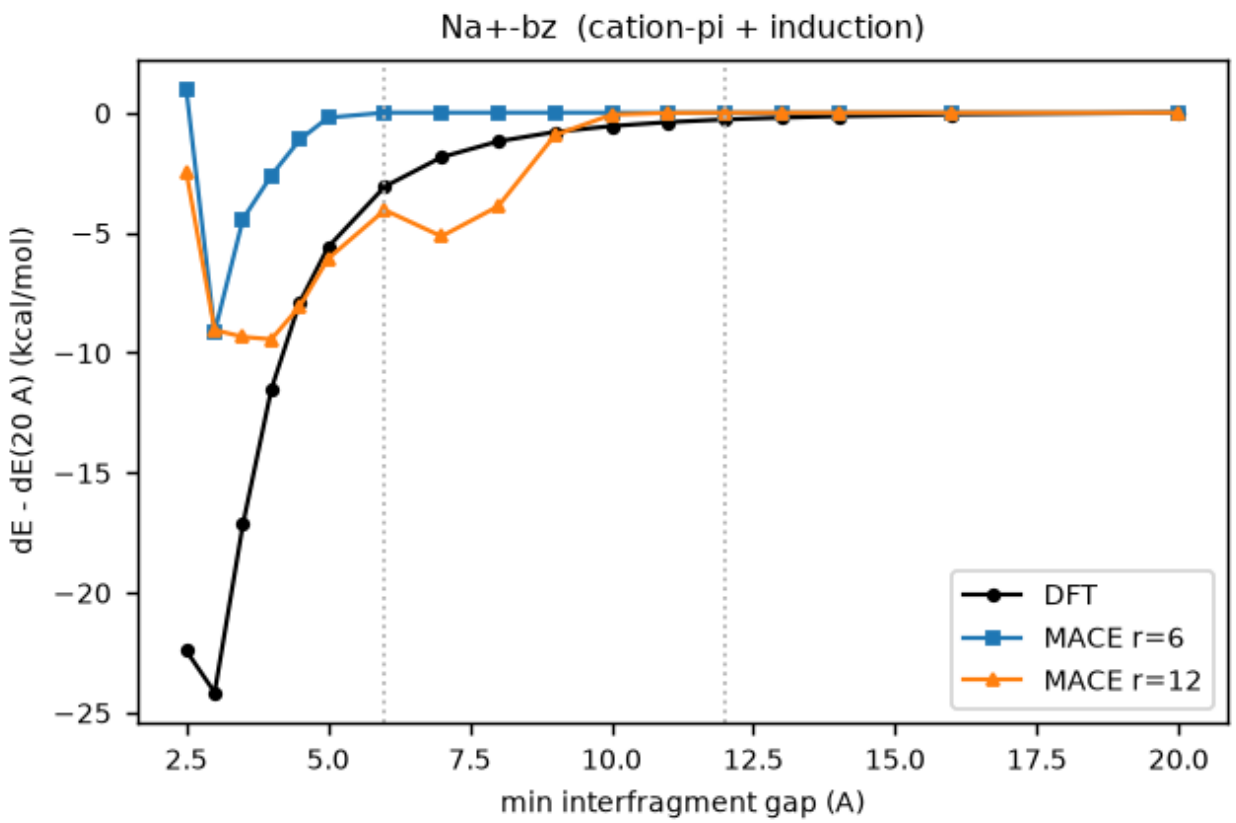


Table 8

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−22.43	+0.96	−2.47	+19.96
3.00	−24.17	−9.14	−9.06	+15.11
3.50	−17.14	−4.43	−9.35	+7.79
4.00	−11.53	−2.62	−9.45	+2.08

4.50	−7.91	−1.13	−8.07	−0.17
5.00	−5.61	−0.21	−6.09	−0.48
6.00	−3.08	+0.00	−4.06	−0.98
7.00	−1.86	+0.00	−5.16	−3.30
8.00	−1.20	+0.00	−3.90	−2.71
9.00	−0.81	+0.00	−0.94	−0.13
10.00	−0.56	+0.00	−0.09	+0.47
11.00	−0.40	+0.00	−0.01	+0.39
12.00	−0.28	+0.00	+0.00	+0.28
13.00	−0.21	+0.00	+0.00	+0.21
14.00	−0.15	+0.00	+0.00	+0.15
16.00	−0.07	+0.00	+0.00	+0.07
20.00	+0.00	+0.00	+0.00	+0.00

Neutral–neutral

Water dimer, aligned dipoles (dipole–dipole, r^{-3}): 12 Å worse

Both models match DFT at 3 Å. Beyond 5 Å DFT decays smoothly, to −0.14 at 8 Å. The 12 Å model holds a plateau near −1 kcal/mol out to 8 Å before dropping. The 6 Å model is flat, which is closer to DFT.

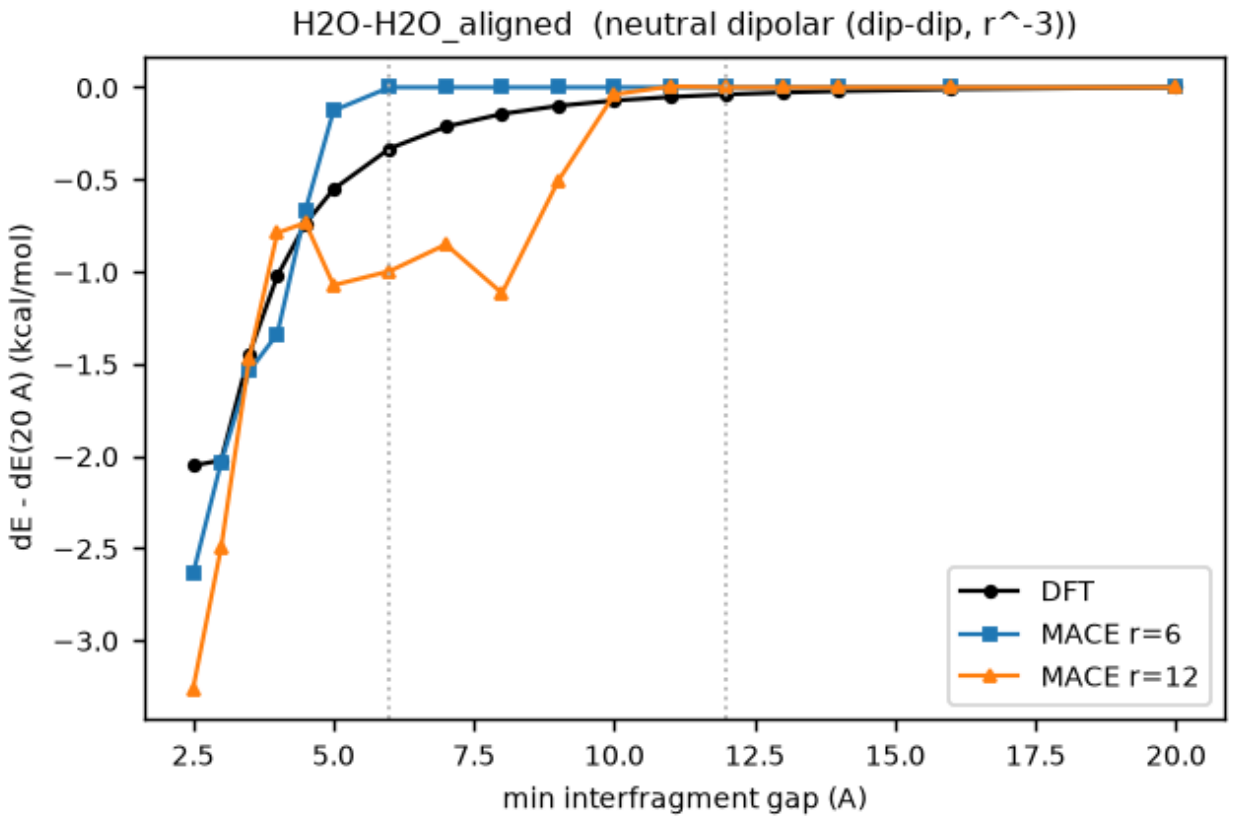


Table 9

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−2.05	−2.64	−3.26	−1.21
3.00	−2.02	−2.03	−2.50	−0.47
3.50	−1.45	−1.54	−1.47	−0.02
4.00	−1.02	−1.34	−0.79	+0.23
4.50	−0.74	−0.67	−0.73	+0.01
5.00	−0.55	−0.13	−1.07	−0.52
6.00	−0.33	+0.00	−1.00	−0.67
7.00	−0.21	+0.00	−0.85	−0.64
8.00	−0.14	+0.00	−1.12	−0.97
9.00	−0.10	+0.00	−0.51	−0.41
10.00	−0.07	+0.00	−0.04	+0.03
11.00	−0.05	+0.00	+0.01	+0.06
12.00	−0.04	+0.00	+0.00	+0.04
13.00	−0.03	+0.00	+0.00	+0.03
14.00	−0.02	+0.00	+0.00	+0.02
16.00	−0.01	+0.00	+0.00	+0.01
20.00	+0.00	+0.00	+0.00	+0.00

Acetonitrile dimer, head to tail (dipole–dipole, r^{-3}): 12 Å worse

The 6 Å model is close to DFT at 3–4 Å. The 12 Å model is 3–5 times too attractive from 3 to 8 Å (−10.5 at 3 Å against DFT −3.0).

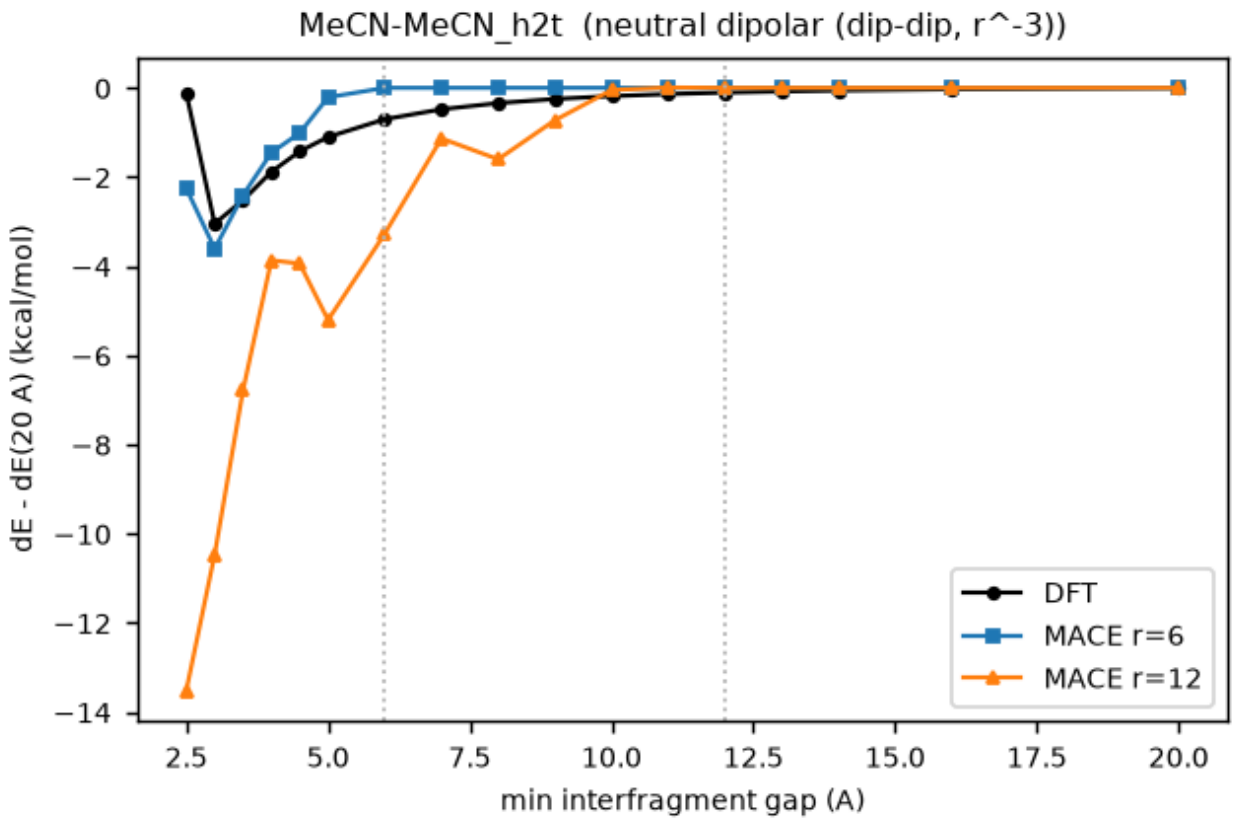


Table 10

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−0.12	−2.25	−13.52	−13.39
3.00	−3.04	−3.61	−10.47	−7.42
3.50	−2.52	−2.41	−6.78	−4.26
4.00	−1.88	−1.45	−3.87	−1.98
4.50	−1.42	−1.01	−3.94	−2.52
5.00	−1.10	−0.21	−5.21	−4.11
6.00	−0.70	+0.00	−3.27	−2.57
7.00	−0.48	+0.00	−1.14	−0.65
8.00	−0.34	+0.00	−1.60	−1.26
9.00	−0.25	+0.00	−0.73	−0.48
10.00	−0.18	+0.00	−0.05	+0.14
11.00	−0.14	+0.00	+0.01	+0.15
12.00	−0.10	+0.00	+0.00	+0.10
13.00	−0.08	+0.00	+0.00	+0.08
14.00	−0.06	+0.00	+0.00	+0.06
16.00	−0.03	+0.00	+0.00	+0.03

20.00	+0.00	+0.00	+0.00	+0.00

Methane dimer (dispersion, r^{-6}): 12 Å worse

DFT is essentially zero beyond 4 Å. The 12 Å model puts in a spurious −2 kcal/mol well at 5–6 Å that fades by 10 Å. The 6 Å model is flat from 5 Å.

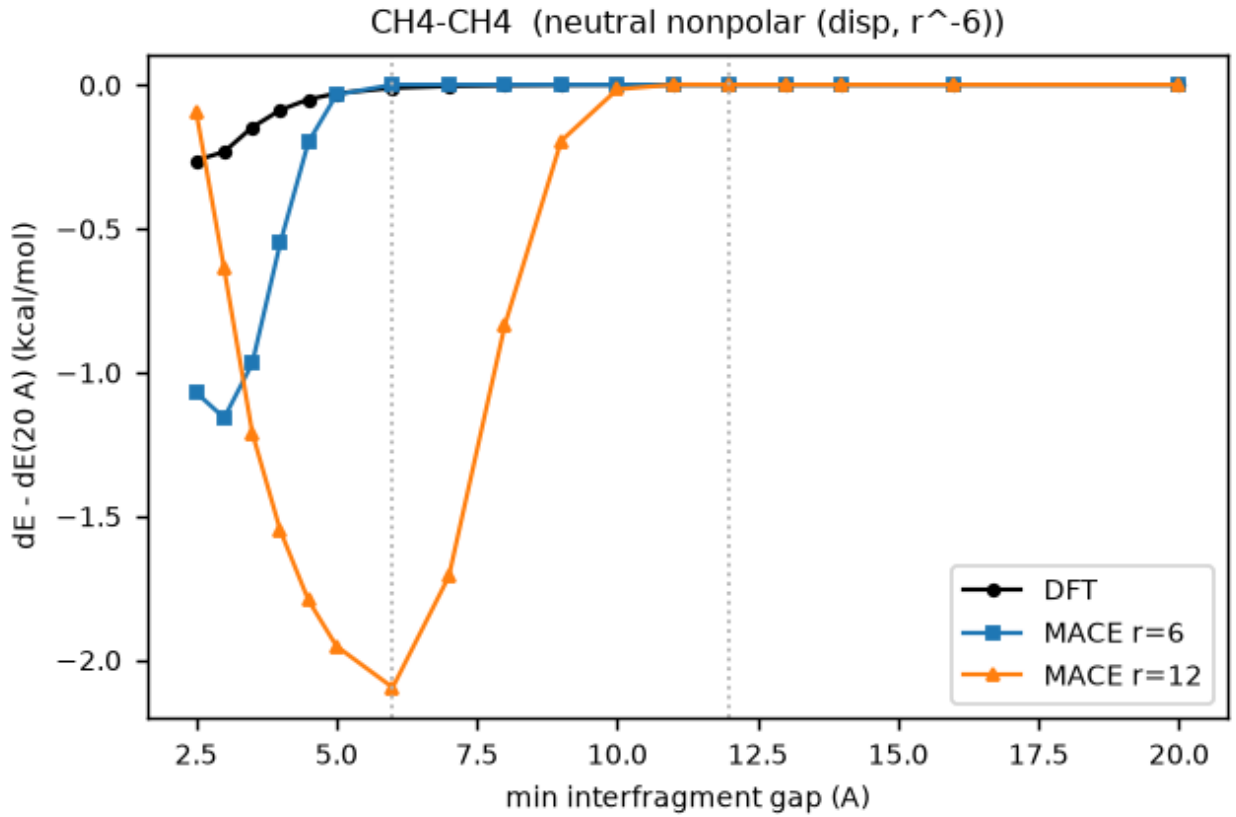


Table 11

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	−0.26	−1.07	−0.09	+0.17
3.00	−0.23	−1.16	−0.64	−0.41
3.50	−0.15	−0.97	−1.21	−1.06
4.00	−0.09	−0.55	−1.55	−1.46
4.50	−0.05	−0.20	−1.79	−1.74
5.00	−0.03	−0.04	−1.95	−1.92
6.00	−0.01	+0.00	−2.10	−2.09
7.00	−0.01	+0.00	−1.71	−1.71
8.00	−0.00	+0.00	−0.83	−0.83
9.00	−0.00	+0.00	−0.20	−0.19
10.00	−0.00	+0.00	−0.02	−0.02
11.00	−0.00	+0.00	+0.00	+0.00

12.00	−0.00	+0.00	+0.00	+0.00
13.00	−0.00	+0.00	+0.00	+0.00
14.00	−0.00	+0.00	+0.00	+0.00
16.00	−0.00	+0.00	+0.00	+0.00
20.00	+0.00	+0.00	+0.00	+0.00

Benzene dimer, stacked (dispersion and quadrupole): 12 Å worse

The 6 Å model reproduces the 3 Å repulsion (+11.0 against DFT +11.7). The 12 Å model misses it (+1.8) and has a spurious −15 kcal/mol well at 6 Å that persists to 10 Å. This is the largest artifact in the set.

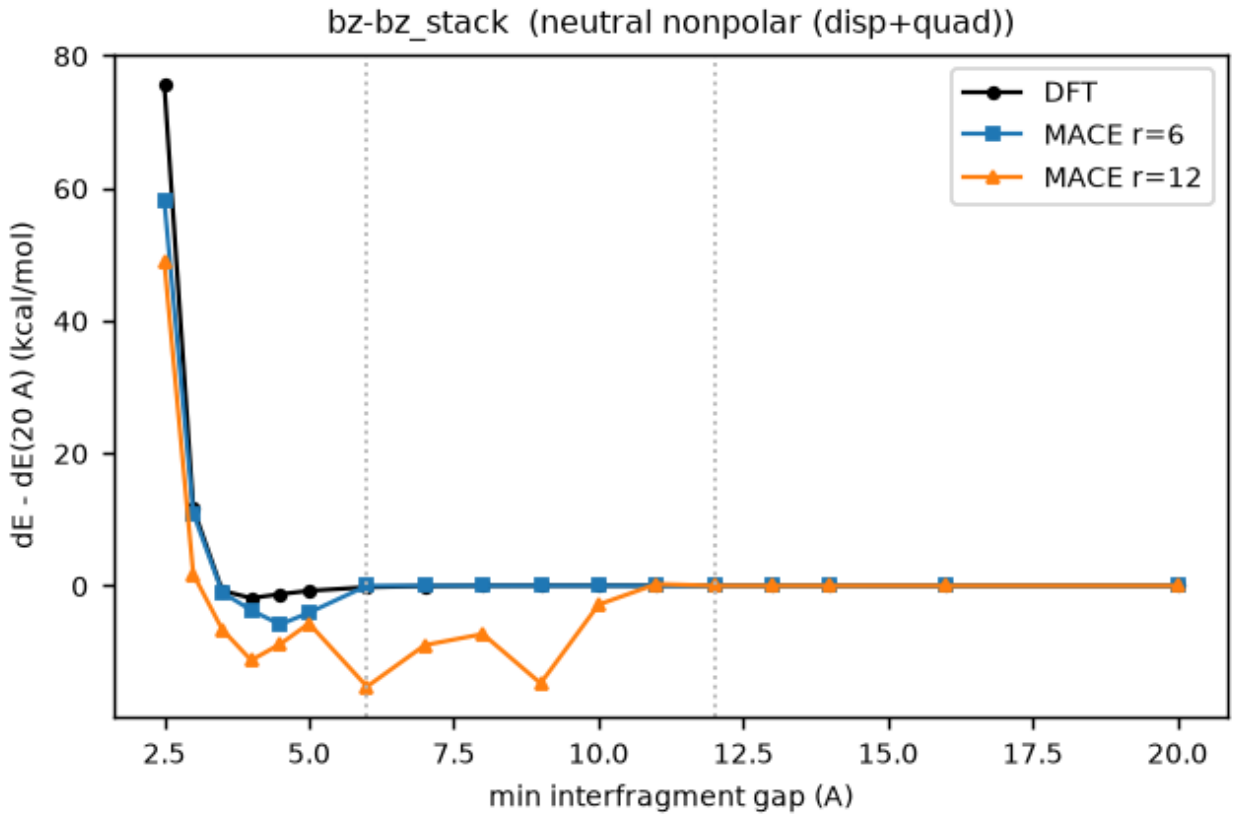


Table 12

gap (Å)	DFT	6 Å	12 Å	12 Å − DFT
2.50	+75.66	+58.09	+48.90	−26.76
3.00	+11.68	+10.96	+1.75	−9.93
3.50	−0.76	−1.02	−6.60	−5.84
4.00	−1.86	−3.69	−11.26	−9.40
4.50	−1.28	−5.85	−8.82	−7.54
5.00	−0.77	−4.07	−5.71	−4.94
6.00	−0.23	+0.00	−15.23	−15.01
7.00	−0.05	+0.00	−8.96	−8.92

8.00	+0.01	+0.00	−7.29	−7.30
9.00	+0.02	+0.00	−14.73	−14.75
10.00	+0.02	+0.00	−2.86	−2.88
11.00	+0.02	+0.00	+0.25	+0.23
12.00	+0.01	+0.00	+0.00	−0.01
13.00	+0.01	+0.00	+0.00	−0.01
14.00	+0.01	+0.00	+0.00	−0.01
16.00	+0.00	+0.00	+0.00	−0.00
20.00	+0.00	+0.00	+0.00	+0.00

Data: `longrange/scan/results/scan_table.tsv` (raw and shifted, all points); ORCA outputs in `longrange/scan/orca/`.